

Project Initialization and Planning Phase

Date	
Team ID	
Project Title	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposal report aims to transform loan approval using machine learning, boosting efficiency and accuracy. It tackles system inefficiencies, promising better operations, reduced risks, and happier customers. Key features include a machine learning-based credit model and real-time decision-making.

Project Overview	
Objective	The primary objective is to develop an ML-powered crop recommendation system that analyzes soil nutrients (N, P, K) and environmental parameters (temperature, humidity, pH, rainfall) to predict the most suitable crop, enabling data-driven agricultural decisions and maximizing yield.
Scope	The project encompasses the full pipeline: dataset generation, exploratory data analysis, training and comparison of multiple classifiers (Logistic Regression, KNN, Decision Tree, Random Forest), prediction and suitability assessment, and a web-based dashboard for researchers and policymakers.
Problem Statement	
Description	Smallholder farmers lack access to data-driven decision-making tools, leading to suboptimal crop selection, soil degradation from monocropping, wasted input costs, and vulnerability to climate variability. Existing solutions are either too expensive (lab soil

	testing), too slow (government extension services), or not data-driven (traditional knowledge passed down generations).
Impact	Solving these issues will result in higher crop yields, reduced input waste, improved soil health through informed crop rotation, and greater financial stability for farmers. For policymakers, it provides data-backed insights for agricultural planning and food security.
Proposed Solution	
Approach	Employing supervised machine learning (Random Forest classifier with 99.5% accuracy) to analyze 7 soil and environmental features and recommend the optimal crop. The system also provides suitability analysis and an exploratory K-Means clustering visualization.
Key Features	- ML-based crop recommendation using 7 soil & environmental parameters - Suitability assessment with soil health & environmental compatibility insights - Comparative evaluation of 4 classifiers (LR, KNN, DT, RF) - Interactive research dashboard with 9 EDA charts + K-Means cluster visualization - REST API for external integration - Web interface accessible on any device (desktop, tablet, mobile)

Resource Requirements

Resource Type	Description	Specification/Allocation
Hardware		
Computing Resources	CPU/GPU specifications, number of cores	CPU: 4+ cores GPU: T4 (optional for training)
Memory	RAM specifications	8 GB
Storage	Disk space for data, models, and logs	50 TB SSD

Software		
Frameworks	Python frameworks	Flask
Libraries	Additional libraries	scikit-learn 1.9, pandas 3.x, numpy 2.x, matplotlib, seaborn, joblib
Development Environment	IDE	Jupyter Notebook, pycharm
Data		
Data	Source, size, format	Kaggle dataset, 614, csv crop_recommendation dataset, 690, csv